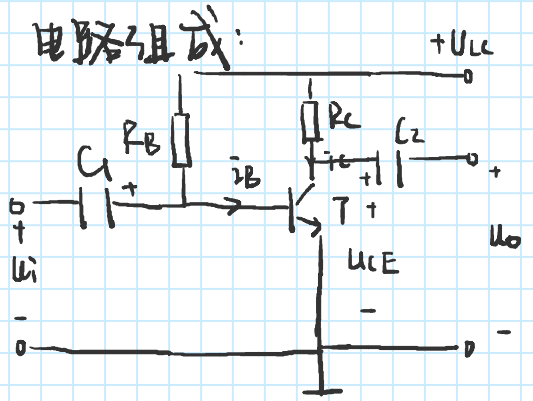
第三章 分立元件基本电路

2023年2月11日 19:16

3.1 共发射极放大电路

3.1.1 电路组成:



3.1.2 静态分析

输入电路: $U_{BE}=U_{CC}-R_BI_B$ $\qquad I_B=\dfrac{U_{CC}-U_{BE}}{R_B}$

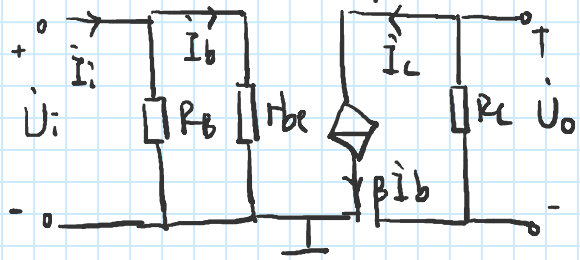
输出电路: $U_{CE}=U_{CC}-R_CI_C$ $\qquad I_C=\beta I_B$

3.1.3 动态分析:

$u_{BE}=U_{BE}+u_i$ $\qquad i_B=I_B+i_b$

$i_C=I_C+i_c$ $\qquad u_{CE}=U_{CE}+u_{ce}$ $\qquad u_o=u_{ce}=-R_Ci_c$ → 反相

2. 微变等效电路分析:



$$A_u=\frac{\Delta U_o}{\Delta U_i}=\frac{\dot U_o}{\dot U_i}=\frac{-\beta R_c}{r_{be}}$$

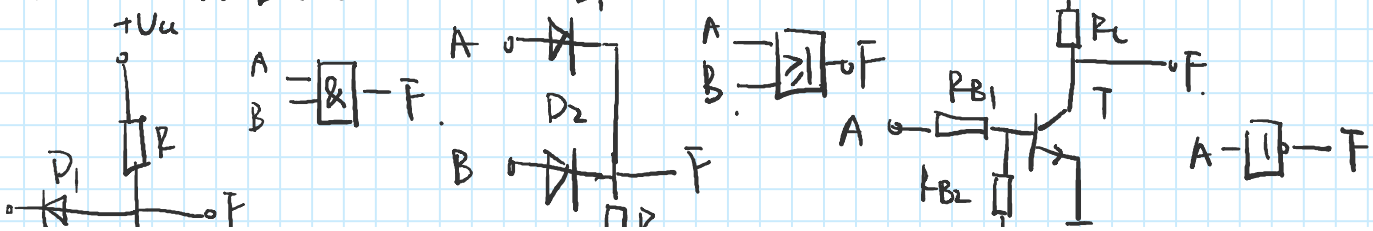
$$r_i=\frac{\dot U_i}{\dot I_i} \qquad \dot U_i=\frac{r_i}{R_s+r_i}\dot U_s$$

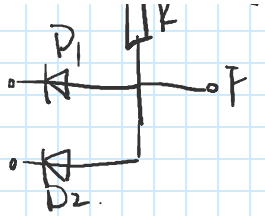
$$r_i=R_B\parallel r_{be}=\frac{R_Br_{be}}{R_B+r_{be}} \qquad r_o=R_C$$

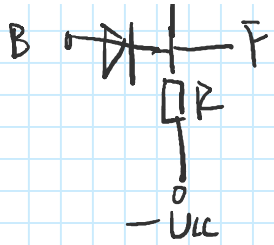
3.1.4 静态工作点的稳定

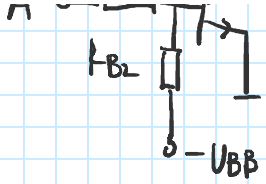
3.4. 分立元件组成的基本门电路

3.4.1 二极管与门电路




D1
K
F
D2


B
F
R
-UCC


RB2
-UBB

A
F